Business Problem Document

Retail Sales Performance & Profitability Analysis | Excel + SQL + Power BI

1. Background

A multi-category retailer in India sells Furniture, Office Supplies and Technology products to Consumer, Corporate and Home Office customers across five regions (North, South, East, West, Central). Between January 2023 and December 2025 sales grew every year, but the profit margin stayed flat at roughly 14%. Leadership suspects that growth is being bought with discounts and wants to know exactly where profit is being lost.

2. Business problem statement

Revenue is growing but profitability is not. Management cannot currently see which products, regions, customers or discount levels create or destroy profit, and has no single view of how performance varies by season or by delivery speed. The business needs a repeatable analysis that identifies profit leaks and recommends specific actions.

3. Objectives

- Build a clean, trusted dataset from messy order exports (dates stored as text, duplicates, inconsistent text, invalid rows).
- Measure sales, profit and margin by year, category, sub-category, region, segment and customer.
- Quantify the effect of discounting on profit and find the discount level at which orders become loss-making.
- Identify loss-making products and weak regions and explain why they underperform.
- Segment customers (RFM) and measure profit concentration.
- Deliver an Excel report, a SQL query library and an interactive Power BI dashboard, plus a recommendations list.

4. Key business questions

#	Question	Answered in
1	What are total sales, profit, margin, orders and average order value? How are they trending year over year?	Excel Summary, SQL Q1-Q2, Power BI page 1
2	Which categories and sub-categories make or lose money?	Excel Summary 2-3, SQL Q4, Power BI page 2
3	At what discount level does profit turn negative?	Excel Summary 6, SQL Q6, Q15, Power BI page 2
4	Which regions underperform and why?	Excel Summary 4 & 8, SQL Q5, Q10, Power BI page 3
5	Is there seasonality we should plan for?	Excel Summary 9, SQL Q12, Power BI page 1
6	Who are the most valuable customers and how concentrated is profit?	Excel Customers, SQL Q8-Q9, Power BI page 3
7	Are delivery times acceptable in every region?	Excel Summary 7-8, SQL Q10

5. Data description

One flat export of order lines (one row per product per order). The dataset is **synthetic**, generated with a fixed random seed to resemble real retail order data and to include realistic data-quality problems. Period: 01-Jan-2023 to 31-Dec-2025. Currency: INR.

Column	Meaning
Order ID, Order Date, Ship Date, Ship Mode	Order header information. Ship Mode: Same Day, First Class, Second Class, Standard Class
Customer ID, Customer Name, Segment	Customer identity and segment (Consumer, Corporate, Home Office)
Region, State, City	Customer geography (5 regions, 20 states)
Product ID, Category, Sub-Category, Product Name	Product hierarchy (3 categories, 13 sub-categories, 65 products)
Sales	Net sales in INR after discount = List price x Quantity x (1 - Discount)
Quantity	Units sold on the line
Discount	Discount given, stored as a fraction (0.20 = 20%)
Profit	Sales minus product cost, in INR (negative = loss)

6. Data quality issues found in the raw file

Issue	Rows	Fix
Exact duplicate order lines	~176	Remove duplicates on Order ID + Product ID
Region / Segment / Name casing and stray spaces	~1,000	TRIM + PROPER
Dates stored as text (dd-mm-yyyy)	all	DATEVALUE / Text to Columns
Discount stored as 20 instead of 0.20	13	Divide by 100 when value > 1
Missing City	107	Fill from the customer's other orders
Ship Date earlier than Order Date	25	Remove (impossible)
Quantity = 0	12	Remove (invalid)

7. Approach

- **Clean** in Excel (and reproduced in Python for audit): standardise text, convert dates, fix discount scale, remove invalid rows and duplicates; log every step.
- **Analyse in Excel**: helper columns, named ranges, SUMIFS/COUNTIFS/AVERAGEIFS summary tables, customer ranking, dashboard charts.
- **Analyse in SQL**: normalise into customers, products, orders and order_lines; run data-quality checks; write 15 analytical queries with window functions, CTEs and RFM.
- **Visualise in Power BI**: star schema (fact_sales + dim_customer, dim_product, dim_date), DAX measures, three report pages with slicers and drill-through.
- **Recommend**: convert findings into specific business actions with an estimated impact.

8. Success criteria

- Excel and SQL totals reconcile exactly (Total Sales, Total Profit, Orders).
- Every data-quality check returns zero problem rows after cleaning.
- Each business question has a clear, quantified answer and an action.
- A non-technical stakeholder can use the dashboard without explanation.

9. Assumptions and limitations

- Data is synthetic: findings demonstrate the method, not real company performance.
- Profit is as reported in the source; no overhead allocation is applied.
- Orders are counted as distinct Order IDs; returns are not in the data.
- Discount-profit relationship is observational: it shows correlation in the data, not an experiment.